Statistische und Stochastische grundlagen der Informatik

Universität Stuttgart

Wintersemester 2025/26

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Chapter 1: Stochastic

1 Introduction

1.1 Bertrands Paradox

This paradox poses a nice introduction of how homogenous sampling is key for good analysis

Lets take a circle of radius 1. We trace a line that crosses the circle at some point. What is the possibility of the length of the inscribed line being $l > \sqrt{3}$?

Note: $\sqrt{3}$ is the length of the size of the maximum possible triangle inscribed in the circle.

1.1.1 First Computation:

We can imagine that the line is always paralel to the y-Axis by just rotating the axis. The intersection with the x-Axis must then be between -1 and 1. For $l = \sqrt{3}$ is $x_0 = \pm\frac{1}{2}$. So the possibility is $|x_0| > \frac{1}{2} = \frac{1}{2}$

1.1.2 Second Computation:

Lets fix one of the intersection points with the circle p_1. The angle between the line and the tangent of the circle p_1 will be called Θ, which will be between 0 and π radians. For $l = \sqrt{3}$, $\Theta = \frac{\pi}{2} \pm \frac{\pi}{6}$ so the possibility of $l > \sqrt{3}$ is $\frac{\pi}{3} < \Theta < \frac{2\pi}{3} = \frac{1}{3}$

1.1.3 Third Computation:

Pick a point $p_m \neq (0, 0)$ uniformly at random somewhere inside the circle. Now consider the chord that is perpendicular to the line connecting p_m with the origin. This construction gives a unique chord for each interior point, and its length depends only on the distance of p_m from the center.

The chord is longer than $\sqrt{3}$ exactly when the point p_m lies closer than $\frac{1}{2}$ to the origin (this follows from the same geometry used in the first computation).

So the probability is simply the ratio of the area of the circle of radius 1/2 to the area of the full unit circle:

$$\frac{\pi(1/2)^2}{\pi(1)^2} = \frac{1}{4}.$$

2 Probability Spaces

A probability space (or *P-space*) consists of the following components:

1. **Sample Space** Ω: A non-empty set of all possible outcomes of a random experiment (e.g., the face values of a die, or the chords considered in Example 2.1).
2. **Event System** $\mathcal{A}$: A collection of subsets of Ω. The sets $A \in \mathcal{A}$ are called *events* (e.g., all even die rolls, or all chords longer than $\sqrt{3}$).
3. **Probability Measure** $P : \mathcal{A} \to \mathbb{R}$: Assigns a probability to every event.

The model does *not* contain "randomness" itself. In probability theory we study properties of a given probability measure P, typically using tools such as random variables and expectations, often combined with combinatorics. Where the probability distribution P comes from, and what its results mean, belongs to the *modeling* outside of probability theory.

Events are sets, so all usual set operations also apply here. In the context of events, they are interpreted as follows:

- $A \cap B$: both events A and B occur

- $A \cup B$: at least one of the events A or B occurs
- $B \setminus A$: B occurs but A does not
- $A^c := \Omega \setminus A$: A does not occur

Later we will require that the event system $\mathcal{A}$ is closed under these operations, so the result of any such operation is again an event. We also require that $\varnothing$ and Ω are included in $\mathcal{A}$. Here, $\varnothing$ is the *impossible* event, and Ω is the *certain* event. Events of the form $\{\omega\}$ with $\omega \in \Omega$ are called *elementary events*.

Definition 1 (Incompatibility)**.** *Two events A and B are called* incompatible *(or* disjoint*) if $A \cap B = \varnothing$.*

For unions of disjoint sets we introduce a summation notation.

Definition 2. *For pairwise disjoint sets $A_1, \ldots, A_n$ (i.e. $A_i \cap A_j = \varnothing$ for $i \neq j$), define:*

$$A_1 + A_2 := A_1 \cup A_2,$$

and more generally

$$\sum_{i=1}^{n} A_i := \bigcup_{i=1}^{n} A_i.$$

3 Properties of Probability Computation

$$\mathbb{P}(\emptyset) = 0$$

$$\mathbb{P}(\Omega) = 1$$

$$\forall A \subset \Omega : \mathbb{P}(A) \geq 0, \quad \mathbb{P}(A) \leq 1$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) \quad \text{for disjoint } A, B.$$

For independent events:

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \mathbb{P}(A_i).$$

4 Random Variables

A key tool for describing probability spaces are *random variables*.

Definition 3 (Random Variable)**.** *A random variable on a probability space is a function*

$$X : \Omega \to \mathbb{R}.$$

(We will slightly refine this definition later in Section 6.)

Random variables allow us to extract numerical properties of outcomes. For example, when rolling a die twice, the sample space is

$$\Omega = \{1, \ldots, 6\}^2,$$

and the sum of both dice is the random variable

$$X(w_1, w_2) := w_1 + w_2.$$

Often we are interested in the preimage of a set $A \subset \mathbb{R}$:

$$X^{-1}(A) := \{\omega \in \Omega : X(\omega) \in A\}.$$

For the example above,

$$X^{-1}([3,4]) = \{(1,2),(1,3),(2,1),(2,2),(3,1)\}.$$

It is common (though somewhat sloppy) to write these sets in shorthand, e.g.

$$\{3 \leq X \leq 4\} := X^{-1}([3,4]).$$

For the probability space we are considering here, the following holds:

$$\{3 \leq X \leq 4\} = \{X = 3\} \cup \{X = 4\}.$$

A random variable always maps into $\mathbb{R}$, so all normal algebraic operations apply, and we may form expressions such as $|X|$ or $\max(X, Y)$.

5 Finite Probability Spaces

5.0.1 Definition.

A finite probability space is a pair $(\Omega, \mathbb{P})$ with

$$\begin{cases} \forall A \subset \Omega : \mathbb{P}(A) \geq 0, \\ \mathbb{P}(\Omega) = 1, \\ \forall A, B \subset \Omega,\ A \cap B = \emptyset : \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B). \end{cases}$$

Because we can write an event A as

$$A = \sum_{w \in A} \{w\},$$

it is enough to define $\mathbb{P}$ on singletons via a probability function

$$p : \Omega \to [0,1],$$

with

$$\sum_{w \in \Omega} p(w) = 1, \qquad \mathbb{P}(A) = \sum_{w \in A} p(w).$$

5.1 Probability Function of a Random Variable

We define

$$\mathbb{P}^X : \mathcal{P}(X(\Omega)) \to \mathbb{R}, \qquad \mathbb{P}^X(B) := \mathbb{P}(X^{-1}(B)).$$

$$\mathbb{P}(X = x) := \mathbb{P}(X^{-1}(\{x\})), \qquad \mathbb{P}(a \leq X \leq b) := \mathbb{P}(X^{-1}([a,b])).$$

5.2 Distribution Function of a Random Variable

$$F^X : \mathbb{R} \to \mathbb{R}, \qquad F^X(x) := \mathbb{P}(X \leq x).$$

With the properties:

$$\lim_{x \to -\infty} F^X(x) = 0, \qquad \lim_{x \to \infty} F^X(x) = 1.$$

5.2.1 Example.

An example of such a probability space is the throwing of a die.

6 Laplace Experiments

Often all elements $w \in \Omega$ have the same elementary probability:

$$\forall w \in \Omega : p(w) = \frac{1}{|\Omega|}.$$

Thus,

$$\forall A \subset \Omega : \mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

Such probability spaces are called **Laplace probability spaces**, and $\mathbb{P}$ is a discrete uniform distribution.

7 Expected Value

For a random variable X we define the expected value as

$$\mathbb{E}(X) := \sum_{w \in \Omega} X(w)\, \mathbb{P}(\{w\}) = \sum_{x \in X(\Omega)} x\, \mathbb{P}(X = x).$$

7.1 Properties of the Expected Value

7.1.1 Linearity.

$$\mathbb{E}\left(\sum_{i=1}^{n} \alpha_i X_i\right) = \sum_{i=1}^{n} \alpha_i\, \mathbb{E}(X_i).$$

7.1.2 Expected Value of the Indicator Function.

If we define

$$\mathbb{1}_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A, \end{cases}$$

then

$$\mathbb{E}(\mathbb{1}_A) = \mathbb{P}(A).$$

7.1.3 Monotonicity.

$$(\forall w \in \Omega : X(w) \leq Y(w)) \;\Rightarrow\; \mathbb{E}(X) \leq \mathbb{E}(Y).$$

7.1.4 Transformation Rule.

For a random variable X and a function $g : \mathbb{R} \to \mathbb{R}$, let $g(X)(w) := g(X(w))$. Then

$$\mathbb{E}(g(X)) = \sum_{x \in X(\Omega)} g(x)\, \mathbb{P}(X = x).$$

8 Variance

For a finite probability space $(\Omega, \mathbb{P})$ and a random variable X we define

$$\mathbb{V}(X) := \mathbb{E}\big((X - \mathbb{E}(X))^2\big)$$

as the variance of X.

$$\sigma(X) := \sqrt{\mathbb{V}(X)}$$

is called the standard deviation.

Chapter 2: Statistic **ADD TYPES OF STATISTIC**

9 Basic Concepts – German-English Translation

- **Statistische Einheit:** Objects that are analyzed
- **Grundgesamtheit:** Population under study
- **Teilgesamtheit:** Subset of the population
- **Stichprobe:** Sample
- **Merkmal / Variable:** Property we are observing
- **Ausprägung / Wert:** The value of the property for a specific object
- **Kenngröße:** Parameter
- **Urliste:** Raw Data Set. The unconverted data before processing

The properties (*Merkmale*) we analyze can have different characteristics:

- **Discrete:** Countable number of values
- **Continuous (stetig):** All possible values within an interval
- **Quasi-Continuous (quasi-stetig):** Continuous but cannot be measured exactly

These properties also have different scales (*Skala*):

- **Nominalskala:** Names, categories, etc.
- **Ordinalskala:** Values have a defined order, but distances between values are not meaningful
- **Kardinalskala:** Measurable, metric scale

10 Data Collection

10.1 Experiment

Obtaining data by actively interacting with the environment under study. Not always possible due to laws or constraints.

10.2 Erhebung (Data Collection)

Usually surveys or observations. Does not interact directly with the environment, less affected by external constraints.

Multiple types:

- **Primärstatistisch:** Data collected directly from the environment
- **Sekundärstatistisch:** Data collected based on existing primary data
- **Tertiärstatistisch:** Data based on transformed or compressed existing data

Vollerhebung: Data collection on the complete population
Important: Avoid bias.

10.2.1 Vollerhebung vs Stichprobe

When sampling, pay attention to how the sample is drawn.

11 Study Designs (*Studiendesigns*)

- **Querschnittsstudie:** Analysis of multiple objects at a specific time point
- **Zeitreihe:** Analysis of one object over time
- **Längsschnittsstudie:** Analysis of multiple objects over time

12 Univariable Descriptive and Explorative Statistics

12.1 First Parameter (*Kenngröße*) – Frequency

If we define a set of n data points where the property X is observed:

$$x_1, \ldots, x_n$$

Let the possible encountered values of X (ordered) be:

$$a_1, \ldots, a_k, \quad k \leq n$$

We can define the following:

- $h(a_j) = h_j$, frequency of a_j in X. **Absolute Frequency**
- $f(a_j) = f_j = \frac{h_j}{n}$, **Relative Frequency**
- $h_1, \ldots, h_k$, **Absolute Frequency Distribution**, displayed through a histogram
- $f_1, \ldots, f_k$, **Relative Frequency Distribution**

12.1.1 Classification

The number of classes k can be computed as:

$$k = \lfloor \sqrt{n} \rfloor \quad \text{for } n \leq 100$$

or, for larger n:

$$k = \lfloor 10 \cdot \log_{10} n \rfloor$$

12.2 Characterization over form parameters

We can classify a distribution on its amount of peaks, if its symetric and basically the form of the graph

12.3 Characterization over empirical distribution function

ONLY FOR ORDINAL SCALED DATA

We can define the following

Absolute Cumulative Frequency Distribution

$$H(x) = h(a_1) + h(a_2) + .. + h(a_j) = \sum_{i:a_i \leq x} h_i$$

Empirical Distribution Function

$$F(x) = f(a_1) + f(a_2) + .. + f(a_j) = \sum_{i:a_i \leq x} f_i$$

This characterization answers the question of: what percentage of the data is $\leq x$?
Doesnt answer: Where is the center of the data...

12.4 Characterization over Central tendency

12.4.1 Central tendency (Lagemaße)

This concept describes the center of a distribution through a numerical value

12.4.2 Definition. Central tendency:

Its a function

$$L : \mathbb{R}^n \to \mathbb{R}$$

with the following property

$$L(x_1 + a, ..., x_n + a) = L(x_1, ..., x_n) + a \ , \forall a, x_1, ..., x_n \in \mathbb{R}$$

The most used central tendency is the arithmetical mean.

12.4.3 Arithmetical Mean:

Given the image list $x_1, ..., x_n$. We can compute the arithmetical mean as follows

$$\overline{x} = \frac{1}{n}(x_1 + ... + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i$$

12.4.4 Median:

If we define this list as sorted $(x_{(1)}, ...x_{(n)})$, the median is the middle point

$$x_{med} \begin{cases} x_{(\frac{n+1}{2})} & \text{if n odd} \\ \frac{x_{(n/2)} + x_{(n/2+1)}}{2} & \text{else} \end{cases}$$

12.4.5 Mode:

The most common element

12.4.6 Geometrical Middle (midpoint):

Given a list $x_1, ..., x_n$

$$x_{geom} = (x_1 \cdot ... \cdot x_n)^{\frac{1}{n}} = (\prod_{i=1}^{n} x_i)^{\frac{1}{n}}$$

12.4.7 Harmonical Middle:

Given a list $x_1, ..., x_n$

$$x_{harm} = \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \frac{1}{x_i}}$$

12.5 Characterization over Statistical Dispersion

12.5.1 Statistical dispersion (Streuungsmaße)

12.5.2 Definition:

Its a function

$$S : \mathbb{R}^n \to \mathbb{R}$$

with the following property

$$S(x_1 + a, ..., x_n + a) = S(x_1, ..., x_n) \ , \forall a, x_1, ..., x_n \in \mathbb{R}$$

12.5.3 Variation width, sample space width

$:= x_{max} - x_{min}$

12.6 p-Quantil (Percentil)

Given a list of ordered data

$$x_1 \leq x_2 \leq \cdots \leq x_n$$

We can define x_p with $0 < p < 1$ as the value that splits the ordered list from above into two parts so that

$$\frac{|\{x_i : x_i \leq x_p\}|}{n} \geq p$$

and

$$\frac{|\{x_i : x_i \geq x_p\}|}{n} \geq 1 - p$$

Basically: The hypothetical value that splits the list in two pieces, the left piece taking p% of the total values

$$x_p = F^{-1}(p)$$

Its similar to the z quantils but with discrete data

12.7 Interquantilerange (Interquartilsabstand)

We define

$$d_Q = x_{0.75} - x_{0.25}$$

In general, the following 5-Point description is pretty helpful to imagine a distribution

$$x_{min}, x_{0.25}, x_{median}, x_{0.75}, x_{max}$$

12.8 Graphical display

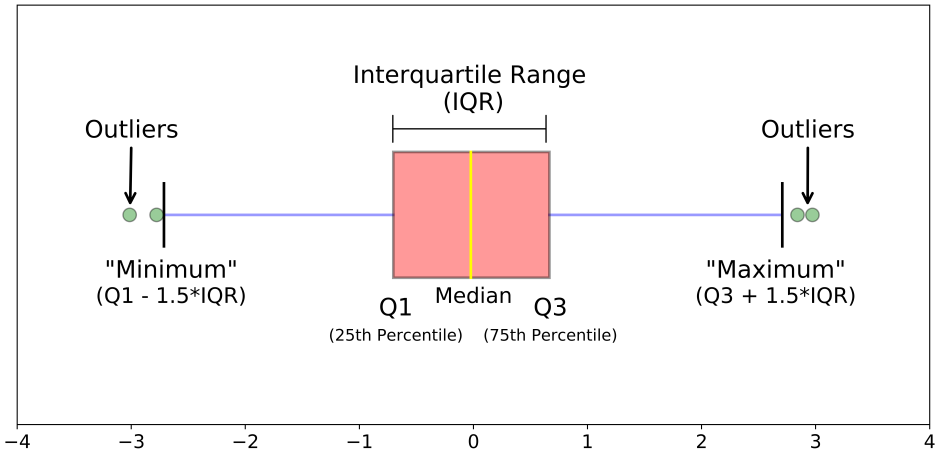
12.8.1 Box Plot:

If we draw all the points of the distribution on a line. The box plot is drawn as follows (image describes both in expanded box plot)

12.8.2 Expanded Box Plot:

Same as box plot just with the fence.

The fence is computed as $\{x_{0.25} - 1.5 \cdot d_Q, x_{0.75} + 1.5 \cdot d_Q\}$ If the fence doesnt land on any point, we **compress** it until the next point on both edges



12.9 Calculation rules

12.9.1 Transformation rule:

For a linear transformation $y = ax + b$ the following stands

$$\tilde{s}_y^2 = a^2 \tilde{s}_x^2$$

Note: $\tilde{s}$ will be described blow

12.9.2 Displacement theorem:

For all $c \in \mathbb{R}$

$$\sum_{i=1}^{n}(x_i - c)^2 = \sum_{i=1}^{n}(x_i - \overline{x})^2 + n(\overline{x} - c)^2$$

For $c = 0$

$$\tilde{s}^2 = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^2\right) - \overline{x}^2$$

12.10 Characterization over Form parameters (v2?)

12.10.1 Skewness (Schiefe)

We can define it as

$$g_m = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^3}{\tilde{s}^3}$$

This explains a bit the simmetry of the data. If < 0 the peak is more on the right, if $= 0$ more symmetric, and if > 0 peak is more on the left

12.10.2 Curvature (Wölbung)

$$\gamma = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^4}{\tilde{s}^4} - 3$$

Talks about the pronunciation of the curvature. $= 0$ represents similarity to $N(0, 1)$, positive means more prominent curve upwards, negative means flatter

12.11 Konzentrationsmaße

Konzentrationsmaße are used to analyze inequality (unequal distribution) of values within a dataset.

12.11.1 Lorenzkurve (Lorenz Curve)

Given a raw dataset on a cardinal scale with non-negative values

$$x_1 \leq x_2 \leq \cdots \leq x_n,$$

we define the traverse chain

$$(0,0), (u_1, v_1), \ldots, (u_n, v_n),$$

where

$$u_j = \frac{j}{n} \quad \text{(proportion of individuals)},$$

$$v_j = \frac{\sum_{i=1}^{j} x_i}{\sum_{i=1}^{n} x_i} \quad \text{(cumulative relative sum)}.$$

The Lorenz curve shows the cumulative share of the total value held by the lowest u_j proportion of individuals. Perfect equality corresponds to the diagonal line $v = u$. The stronger the downward deviation from the diagonal, the higher the inequality.

12.11.2 Gini Coefficient

For an ordered dataset, the Gini coefficient is defined as

$$G = \frac{\text{Area between the diagonal and the Lorenz curve}}{\text{Area between the diagonal and the } u\text{-axis}} = 2 \cdot \text{Area between the diagonal and the Lorenz curve.}$$

For discrete data:

$$G = \frac{2\sum_{i=1}^{n} i x_i}{n \sum_{i=1}^{n} x_i} - \frac{n+1}{n}.$$

The Gini coefficient measures inequality: $G = 0$ indicates perfect equality, and larger values indicate stronger inequality. The maximum value depends on the sample size n.

12.11.3 Normalized Gini Coefficient

To remove the dependence on sample size, the Gini coefficient can be normalized:

$$G^* = \frac{G}{G_{\max}} = \frac{n}{n-1} G \in [0,1].$$

13 Multivariate Descriptive and Explorative Statistics

13.1 Contingency Table

A contingency table summarizes the joint distribution of two variables.

Given variables X and Y with observed values

$$x_1, \ldots, x_k \quad \text{and} \quad y_1, \ldots, y_m,$$

the table is:

	x_1	...	x_k	total
y_1				
$\vdots$				
y_m				
total				

The table entries can be:

- Absolute frequencies $h_{i,j}$
- Relative frequencies $f_{i,j}$

The row and column totals (e.g. $h_{\cdot,j} = \sum_{i=1}^{k} h_{i,j}$) are called **marginal frequencies**. Their relative versions are called **relative marginal frequencies**.

13.2 Conditional Frequencies

The conditional frequency of $X = a_i$ given $Y = b_j$ is

$$f_X(a_i \mid b_j) = \frac{h_{i,j}}{h_{\cdot,j}}.$$

13.3 Correlation Analysis in Contingency Tables

13.3.1 χ^2 Coherence Measure

$$\chi^2 = \sum_{i=1}^{k} \sum_{j=1}^{m} \frac{(h_{i,j} - \tilde{h}_{i,j})^2}{\tilde{h}_{i,j}} \in [0, \infty),$$

where the expected frequencies under independence are

$$\tilde{h}_{i,j} := \frac{h_{i,\cdot} h_{\cdot,j}}{n}.$$

Explanation: χ^2 compares observed frequencies with expected frequencies assuming independence. Large values indicate strong deviations from independence, hence statistical dependence between X and Y. If $\chi^2 = 0$, the variables are independent.

13.3.2 Contingency Coefficient

$$K = \sqrt{\frac{\chi^2}{n + \chi^2}},$$

with

$$K \in \left[0, \sqrt{\frac{M-1}{M}}\right], \quad M = \min\{k, m\}.$$

13.3.3 Normalized Contingency Coefficient

$$K^* = \frac{K}{\sqrt{\frac{M-1}{M}}}, \quad K^* \in [0, 1].$$

The normalized coefficient allows comparison across tables of different sizes.

13.3.4 Empirical Correlation Coefficient

Also known as the **Bravais-Pearson correlation coefficient** or **Pearson's correlation coefficient**.

For a sample $(x_1, y_1), \ldots, (x_n, y_n)$, we define the empirical correlation coefficient as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2} \cdot \sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}}$$

where $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ are the sample means.

This can be equivalently written using sample covariance and standard deviations:

$$r = \frac{\tilde{s}_{xy}}{\tilde{s}_x \cdot \tilde{s}_y}$$

where $\tilde{s}_{xy} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$ is the sample covariance, and $\tilde{s}_x = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2}$ and $\tilde{s}_y = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \overline{y})^2}$ are the sample standard deviations.

Properties:

- $r \in [-1, 1]$ (the correlation coefficient is bounded)
- $r > 0$ indicates a positive linear correlation
- $r < 0$ indicates a negative linear correlation
- $r = 0$ indicates no linear correlation
- $|r| = 1$ indicates perfect linear correlation

13.4 Linear Regression

Linear regression is a statistical method for modeling the relationship between a dependent variable Y and one or more independent variables (predictors) X. The goal is to find a linear equation that best describes how these variables relate.

Since real data contains measurement errors and random variation, we model the relationship as:

$$Y = f(X) + \epsilon$$

where ϵ represents the random error term with $\mathbb{E}[\epsilon] = 0$.

For simple linear regression with a single predictor, we assume:

$$f(X) = \alpha + \beta X$$

where α is the intercept and β is the slope parameter.

13.4.1 Method of Least Squares

The method of least squares finds the estimators $\hat{\alpha}$ and $\hat{\beta}$ that minimize the sum of squared residuals. We define the objective function:

$$Q(\alpha, \beta) = \frac{1}{n}\sum_{i=1}^{n}\epsilon_i^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\alpha + \beta x_i))^2$$

Taking partial derivatives with respect to α and β and setting them equal to zero yields the normal equations. The solution is:

$$\hat{\beta} = \frac{\tilde{s}_{xy}}{\tilde{s}_x^2} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

$$\hat{\alpha} = \overline{y} - \hat{\beta}\overline{x}$$

Note that $\hat{\beta} = r \cdot \frac{\tilde{s}_y}{\tilde{s}_x}$, showing the connection between the regression slope and the correlation coefficient.

13.5 Residual Analysis

Residual analysis is a technique to assess how well a regression model fits the observed data. The fitted values are $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$, and the residuals are $e_i = y_i - \hat{y}_i$.

13.5.1 Decomposition of Total Variation

The total variation in the dependent variable can be decomposed as:

$$\underbrace{\text{SST}}_{\text{Total Sum of Squares}} = \underbrace{\text{SSE}}_{\text{Explained Sum of Squares}} + \underbrace{\text{SSR}}_{\text{Residual Sum of Squares}}$$

$$\sum_{i=1}^{n}(y_i - \overline{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

This decomposition states that the total variability in Y equals the variability explained by the model plus the unexplained (residual) variability.

13.5.2 Coefficient of Determination

The coefficient of determination, denoted R^2, measures the proportion of variance in the dependent variable that is explained by the independent variable(s):

$$R^2 = \frac{\text{SSE}}{\text{SST}} = \frac{\sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2} = 1 - \frac{\text{SSR}}{\text{SST}}$$

Properties:

- $R^2 \in [0, 1]$
- $R^2 = 0$ means the model explains none of the variability
- $R^2 = 1$ means the model explains all of the variability (perfect fit)
- For simple linear regression: $R^2 = r^2$ (the square of the correlation coefficient)

13.6 Quantile-Quantile Diagrams (QQ-Plots)

A **QQ-plot** is a graphical tool to compare two distributions by plotting their quantiles against each other.

13.6.1 How to Build a QQ-Plot

- **X-axis:** Quantiles x_p from the first distribution
- **Y-axis:** Quantiles y_p from the second distribution
- Plot points (x_p, y_p) for probabilities $p = 0.00, 0.05, 0.10, \ldots, 1.00$

13.6.2 Normal QQ-Plots

The most common use is comparing your data to a normal distribution $N(0, 1)$. These are called **Normal QQ-plots** or **NQ-plots**.

13.6.3 How to Read QQ-Plots

- **Points close to diagonal line:** Data follows the reference distribution well
- **Points far from diagonal:** Data doesn't match the reference distribution
- **Axis scaling doesn't matter:** Focus on the pattern, not the numbers
- **Quick visual check:** Easy way to see if your data is normally distributed

13.7 Correlation and Causality

Important: High correlation does NOT mean one variable causes the other. Correlation $\neq$ Causation.

Here are common mistakes when interpreting correlations:

13.7.1 Spurious Correlation (Scheinkorrelation)

Two variables are highly correlated but have no real relationship.

Why it happens:

- A hidden third variable affects both
- Pure coincidence
- Both follow the same external trend

Example: Ice cream sales and drowning deaths are correlated. But ice cream doesn't cause drowning—hot weather causes both!

13.7.2 Hidden Correlation (Verdeckte Korrelation)

Overall data shows no correlation, but relationships exist within subgroups.

Why it happens: Combining different groups hides their individual patterns.

Example: Total tobacco consumption stays constant over time (no correlation). But when you separate by gender:

- Men smoke less over time (negative correlation)
- Women smoke more over time (positive correlation)

These opposite trends cancel out in the combined data!

13.7.3 Outliers (Ausreißer)

Extreme data points that don't fit the general pattern.

Why they matter:

- Can make correlation look stronger or weaker than it really is
- Can change the regression line dramatically
- Might be errors or important unusual cases

What to do:

- Always plot your data to spot outliers
- Check if outliers are mistakes or real observations
- See how results change with and without outliers